

1. Administrative & Study Identification

Full Study Title: A Randomized Double-blind, Placebo-controlled, Multicenter Study to Evaluate the Efficacy, Safety and Tolerability of Pelacarsen (TQJ230) in US Black/African American & Hispanic Patient Populations With Elevated Lp(a) and Established Atherosclerotic Cardiovascular Disease **Short Title/ Acronym:** Lp(a) Lowering Study of Pelacarsen (TQJ230) in US Black/African American and Hispanic Participants With Elevated Lp(a) and Established ASCVD **Protocol Number:** CTQJ230A12303 **NCT Number:** NCT06267560 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Pelacarsen (TQJ230) **Phase of Development:** Phase IIIB **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy (measured by the reduction of Lp(a) levels) of pelacarsen (TQJ230) 80 mg s.c. once monthly (QM) compared to placebo in US Black/African American and US Hispanic participants. **Secondary Objectives:** To evaluate the safety and tolerability of pelacarsen (TQJ230) 80 mg s.c. QM compared with placebo.

2.2 Background & Rationale To address the unmet need for targeted pharmacological therapies to lower lipoprotein(a) [Lp(a)] levels in specific high-risk populations. Elevated Lp(a) is an independent, causal risk factor for cardiovascular events. This study is being conducted to evaluate pelacarsen's ability to lower Lp(a) in US Black/African American and Hispanic populations who have established atherosclerotic cardiovascular disease (ASCVD) and are already receiving standard-of-care treatments for other cardiovascular risk factors.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Quadruple masked) **Randomization:** Randomized (Parallel assignment)

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 422 participants **Number of Sites:** Multicenter (~106 locations in the US and Puerto Rico) **Estimated Study Start:** 24-Apr-2024 **Estimated Primary Completion:** 31-Mar-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female US Black/African American and US Hispanic participants 18 to ≤ 80 years of age. 2. Lp(a) ≥ 125 nmol/L at the screening visit, measured at the Central laboratory. 3. On Standard of Care (SoC) therapy for risk factors other than Lp(a), including LDL-C (LDL-C lowering therapy dose stable for at least 30 days), elevated blood pressure and diabetes, at the randomization visit according to local practice/guidelines. 4. Established ASCVD disease defined as documented: Coronary heart disease (CHD) and/or Cerebrovascular disease (CVD) and/or Peripheral arterial disease (PAD).

4.2 Exclusion Criteria 1. Uncontrolled hypertension. 2. Heart failure New York Heart Association (NYHA) class IV. 3. History of malignancy of any organ system. 4. History of hemorrhagic stroke or other major bleeding. 5. Platelet count $< 140,000$ per mm^3 . 6. Active liver disease or hepatic dysfunction. 7. Significant kidney disease. 8. Pregnant or nursing women.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Pelacarsen (TQJ230) 80 mg once monthly (QM) or matching placebo. **Route of Administration:** Subcutaneous (s.c.) injection. **Duration of Treatment:** 52 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of Care (SoC) treatments for cardiovascular risk factors according to local practice/guidelines (e.g., stable LDL-C lowering therapy, anti-hypertensives, diabetes management). **Prohibited:** Concurrent use of other investigational treatments targeting Lp(a) reduction or interfering with lipid metabolism parameters not approved as SoC.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, Baseline Lp(a) Central Labs, SoC stability confirmation
Baseline	Day 0	Randomization, First Dose (s.c. injection)
Interim	Monthly (QM)	Pelacarsen or Placebo administration, Safety Labs, Efficacy Assessment (Lp(a) tracking)
End of Study	Week 52	Final Vital Signs, Exit Interview, IP Accountability, Final Efficacy Blood Draw

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change in log-transformed Lp(a) concentration (or level) after 52 weeks. **Measurement Method:** Central Laboratory Analysis.

7.2 Secondary & Exploratory Endpoints - Evaluation of safety and tolerability profiles over the 52-week study period. - Proportion of participants achieving specific Lp(a) reduction thresholds.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via HCP interview, lab monitoring, and injection site reaction assessments during monthly visits. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours of site awareness). **Data Monitoring Committee (DMC):** Independent safety monitoring mechanisms as established by Novartis standard protocols.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy evaluation; Safety Set for AE/SAE and tolerability outcomes. **Sample Size Justification:** $N = 422$ is appropriately powered to detect significant differences in log-transformed Lp(a) concentration reductions between the pelacarsen and placebo arms. **Handling of Missing Data:** Mixed-Effect Model Repeated Measure (MMRM) or Multiple Imputation as typically applied in lipid-lowering phase III clinical trials.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to track source data verification (SDV) and protocol adherence. **Audit Trail:** Verified electronic Case Report Form (eCRF) locking and rigorous data clarification cycles.

11. Study Identifiers & Governance

Primary ID: CTQJ230A12303 **Registry Number:** NCT06267560 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** Lp(a) Lowering Study of Pelacarsen (TQJ230) in US Black/African American and Hispanic Participants With Elevated Lp(a) and Established ASCVD **Official Regulatory Title:** A Randomized Double-blind, Placebo-controlled, Multicenter Study to Evaluate the

Efficacy, Safety and Tolerability of Pelacarsen (TQJ230) in US Black/African American & Hispanic Patient Populations With Elevated Lp(a) and Established Atherosclerotic Cardiovascular Disease

12. Clinical Framework (ASCVD Specific)

Primary Condition: Atherosclerotic Cardiovascular Disease (ASCVD)
Diagnosis Threshold: Elevated Lp(a) ≥ 125 nmol/L
Medical Methodology: Standard of Care (SoC) combined with targeted Antisense Oligonucleotide (ASO) therapy
Optimization Target: Reduction in serum Lp(a) levels

13. Study Procedures & Methodology

Management Pathway: Background cardiovascular SoC alongside monthly s.c. investigational product administration.
Education Protocol (HCP): Training on protocol specifics, subcutaneous administration techniques, and central laboratory sample handling.
Education Protocol (Patient): CV risk factor management adherence counseling and injection reporting.
Quality Audit Mechanism: Routine sponsor compliance audits and centralized Lp(a) lab quality control.

14. Observation & Follow-Up Schedule

Total Duration: 52 weeks (approx. 1 year of treatment)
Onsite Visit Cadence: Monthly (QM) for investigational product dosing and monitoring
Remote Monitoring Frequency: Periodic continuous remote data review
Checklist Review Interval: Every 4 to 12 weeks for data clarification checks

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				